TQEC 2/25

Presentations

Bella — spatial hadamard
Configurations using spatial junctions
Fixed boundary convention from reivew paper
Need input on PR: suggestions of configurations + feedback on the functionality

Kabir — tqec development updates (2/24 Slide Deck)
YHalCube — good example of where the state of the art on development is
tqecd needs adjustment for specifics of Y transition route
End of stabilizers from spatial arm
Current hybrid approach should be avoided
New test case — hadamards along spatial axis
Reach out to Kabir to learn more about this
Want to upgrade tqecd to handle larger circuits
Sliding surface codes: fast/slow sliding (2d rounds vs d rounds)
Allows you to move multiple qubit patached in parallel — read attached paper
Can lower spacetime volume of a program in walking surface code
Want to integrate slow sliding pipes into
Vision: make topologiq support all block types
Interoperability with PyZX — qasm parsing now supports reset/measurement
Want simple "dead-end" nodes to be used from PyZX
Can provide feedback to slides, shared

Tianyi — tqecd
Major bottlenecks Pycryptosat solver, Pauli string implementation
_find_cover_sat
Gaussian elimination over binary matrices to finish in polynomial time, currenty runs in exponential time
Can use stim's pauli string class, don't need to convert

Sam — moment.py and detector database performance improvements
Replace file writing with bytes.IO
Potential solution to bottlenecks — not call the remapping anymore. Just remap the circuits with correct qubit numberings.
Copy qubit indices, pass down arguemnts.